Hints: Number theory – 4

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Legendre: $v_p(n!) = \sum_{m \leqslant n} v_p(m) = \sum_{i \geqslant 1} \#\{m \leqslant n : p^i \mid m\}$. For the closed form, write $n = \sum c_j p^j$; then $\lfloor n/p^i \rfloor = \sum_{j \geqslant i} c_j p^{j-i}$. Sum over i.

Hint 2. Kummer: $v_p\binom{n}{k} = \frac{s_p(k)+s_p(n-k)-s_p(n)}{p-1}$, and each carry lowers the digit sum by $p-1$.

Hint 3. All $\binom{n}{k}$ are odd if and only if adding k and $n-k$ in base 2 never carries, for every k. That means n has no 0 digit below its top digit: $n = 2^m - 1$. Next, $v_2\binom{2n}{n} = s_2(n)$ (doubling carries once per 1-digit). Finally, $\binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1}\binom{2n}{n}$ is an integer (Catalan).

Problem 2

Hint 1. LTE, odd p: $\frac{a^n-b^n}{a-b} = \sum a^{n-1-i}b^i \equiv na^{n-1} \pmod p$. This proves the case $p \nmid n$. For $n = p$, show that $p \parallel \frac{a^p-b^p}{a-b}$ by writing $b = a + tp$. Induct. For $p = 2$ and even n, factor $a^n - b^n = (a^{n/2} - b^{n/2})(a^{n/2} + b^{n/2})$, or use $a^2 - b^2$ first.

Hint 2. $v_2(3^n - 1)$ is 1 for odd n and $v_2(2) + v_2(4) + v_2(n) - 1 = 2 + v_2(n)$ for even n. Solving $2 + v_2(n) \geqslant n$: answer $n \in \{1, 2, 4\}$.

Hint 3. First make $p \mid a - b$: $\mathrm{ord}_7(2) = 3$ and $3 \mid 2058$, so write $2^{2058} - 1 = 8^{686} - 1^{686}$. LTE gives $v_7 = v_7(8-1) + v_7(686) = 1 + 3 = 4$, since $686 = 2 \cdot 7^3$.

Problem 3

Hint 1. Compare v_p of both sides. Note $v_p(\mathrm{lcm}(1, \ldots, m)) = \#\{j \geqslant 1 : p^j \leqslant m\}$.

Hint 2. $\sum_i \#\{j : p^j \leqslant n/i\} = \sum_j \#\{i : i \leqslant n/p^j\} = \sum_j \lfloor n/p^j \rfloor = v_p(n!)$.

Problem 4

Hint 1. Every prime factor of $n!+1$ exceeds n. Compare with the multinomial coefficient $\frac{(2012n)!}{(n!)^{2012}}$.

Hint 2. $\gcd(n! + 1, n!) = 1$, so $n! + 1$ divides $\frac{(2012n)!}{(n!)^{2012}} \leqslant 2012^{2012n}$. But $n!$ grows faster than any C^n. Answer: the set is finite.

Problem 5

Hint 1. Bound a_n/n. If $n = ab$ with $1 < a < b$, then $n \mid b!$, so $a_n \leqslant b = n/a \leqslant n/2$.

Hint 2. If $n = q^2$ with q prime, then $a_n = 2q \leqslant n/2$ for $q \geqslant 5$. The exceptions are $n = 4$ ($a_4 = 4$) and $n = 9$ ($a_9 = 6$).

Hint 3. So apart from two terms, $(a_n/n)^n \leqslant 2^{-n}$. Answer: the series converges.